



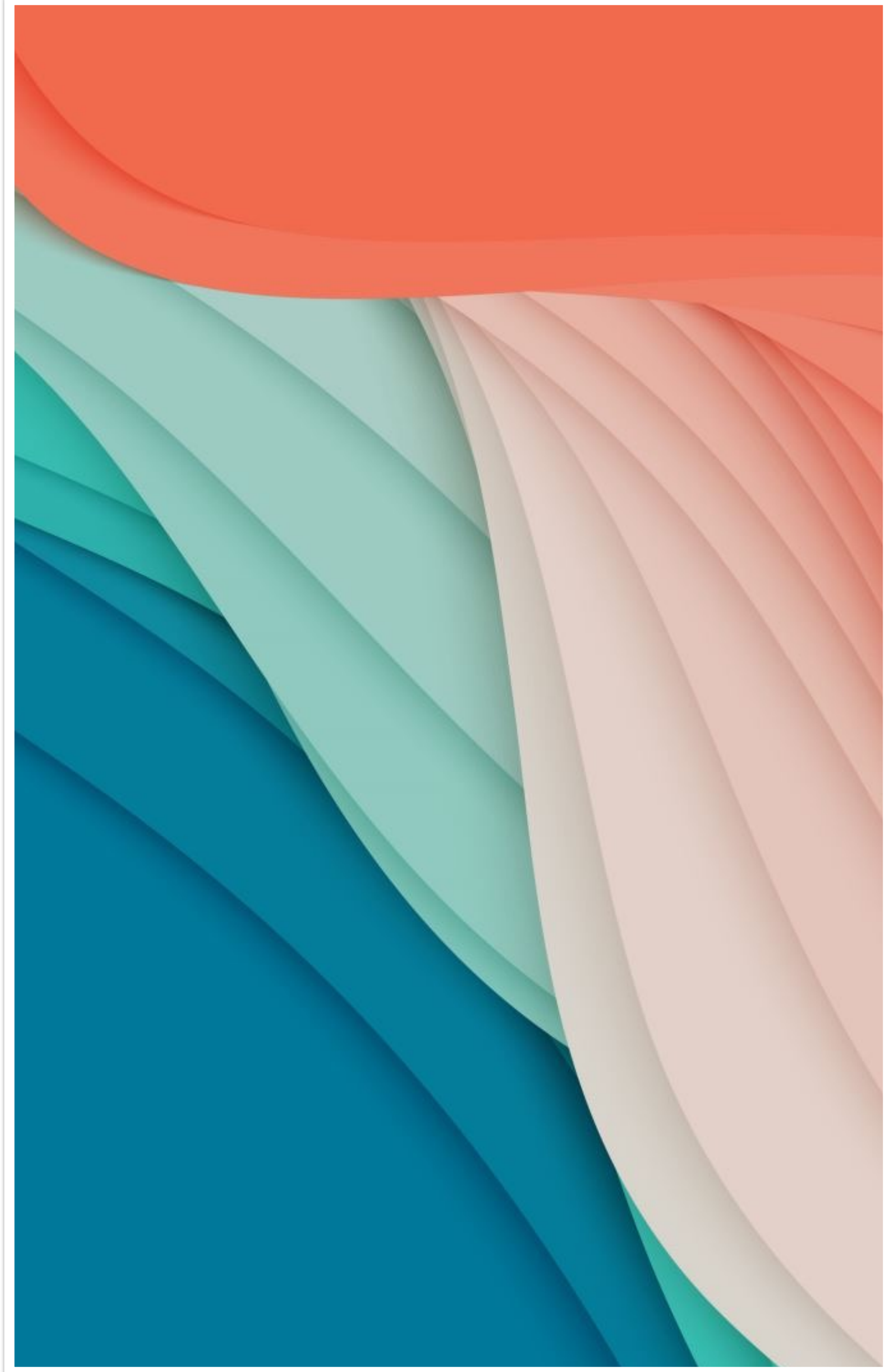
Git - Part I

Introduction & Basics

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Contenu du cours

1. Git et github

- a. Initiation à git : commandes de base.
- b. Utilisation de Visual Studio Code.
- c. Branches dans git.
- d. Collaboration dans git (GitHub).

2. Docker

- a. Bases de docker.
- b. Utilisation de docker hub.
- c. Automatisation avec docker-compose.



Projet



Projet final



Projet final 2



Projet final 3



Projet final final



Projet final finaal



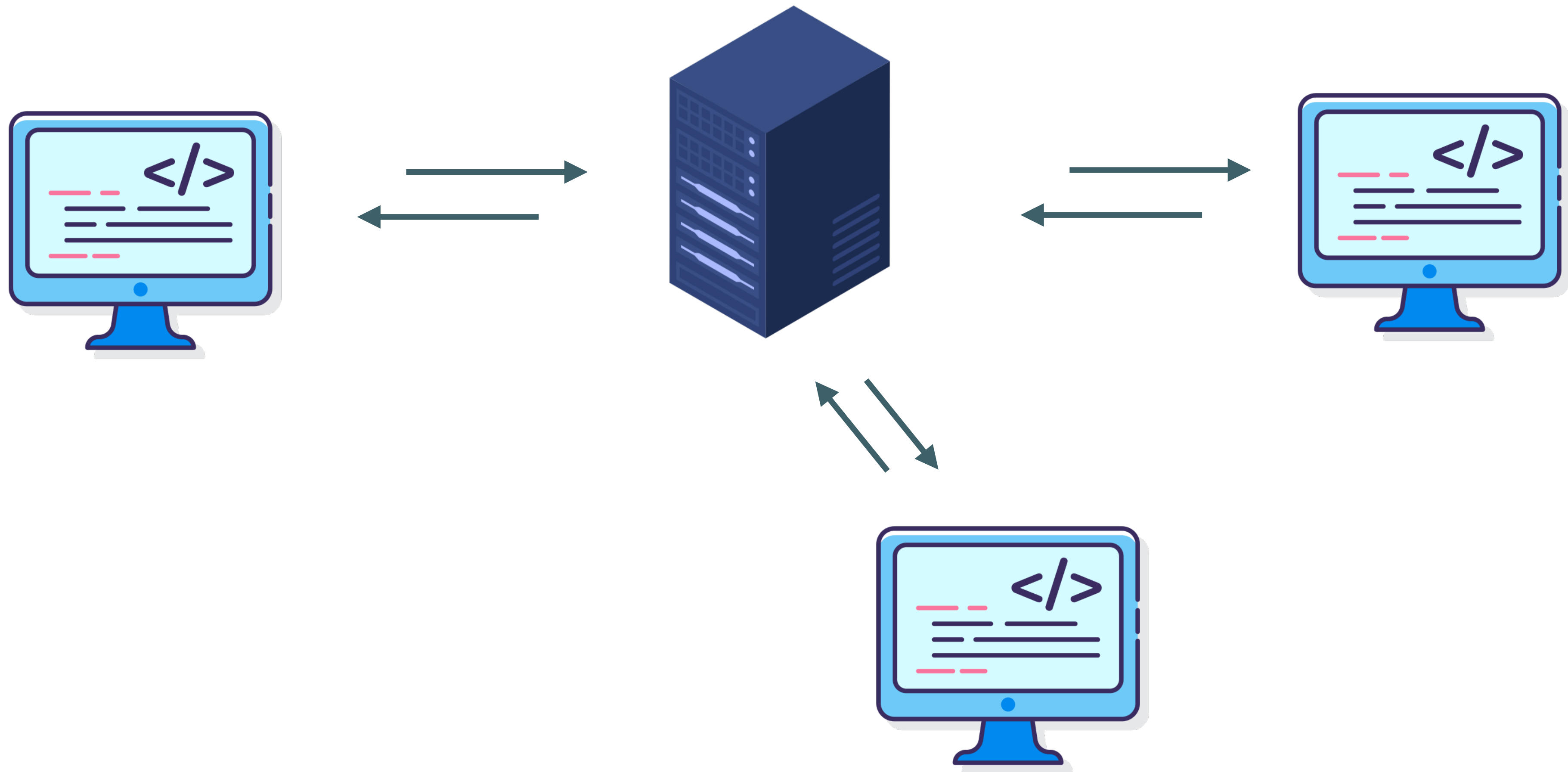
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What is Git ?

- ◆ Git is a **distributed** version control system (VCS).
- ◆ Track changes made on files and revert back to them (like a time traveling machine).
- ◆ Developed and released in 2005 by Linus Torvalds, the creator of Linux.
- ◆ It is a free and open source.
- ◆ According to a [Stack Overflow developer survey](#) over 87% of developers use Git.

What is a distributed VCS

No single point of failure

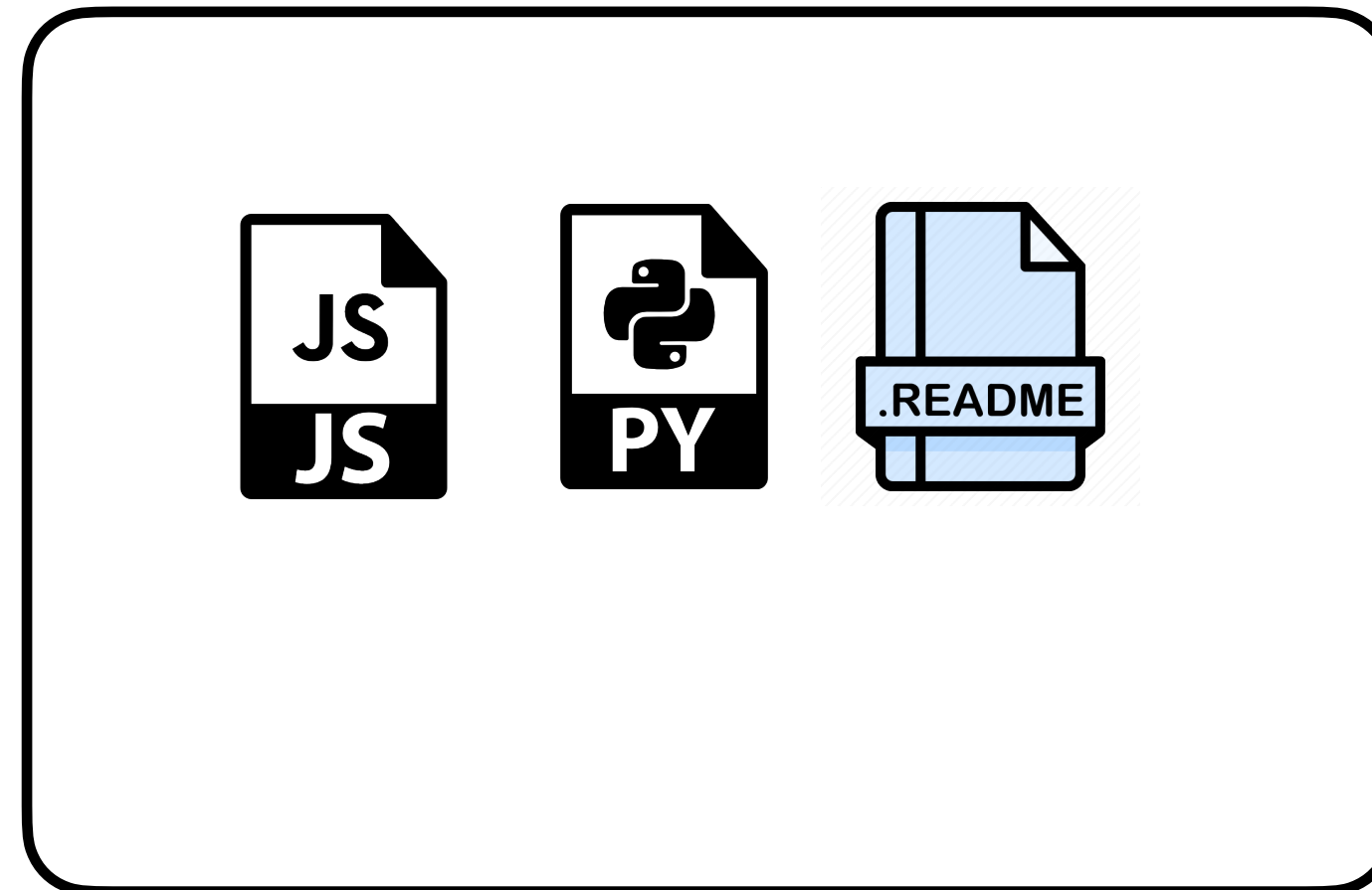


Git hosting service

- ◆ Web hosting service that hosts git projects on the cloud.
- ◆ The most used ones are : Github, bitbucket and gitlab.
- ◆ **Github** is the most popular one, it was bought by Microsoft in 2018.

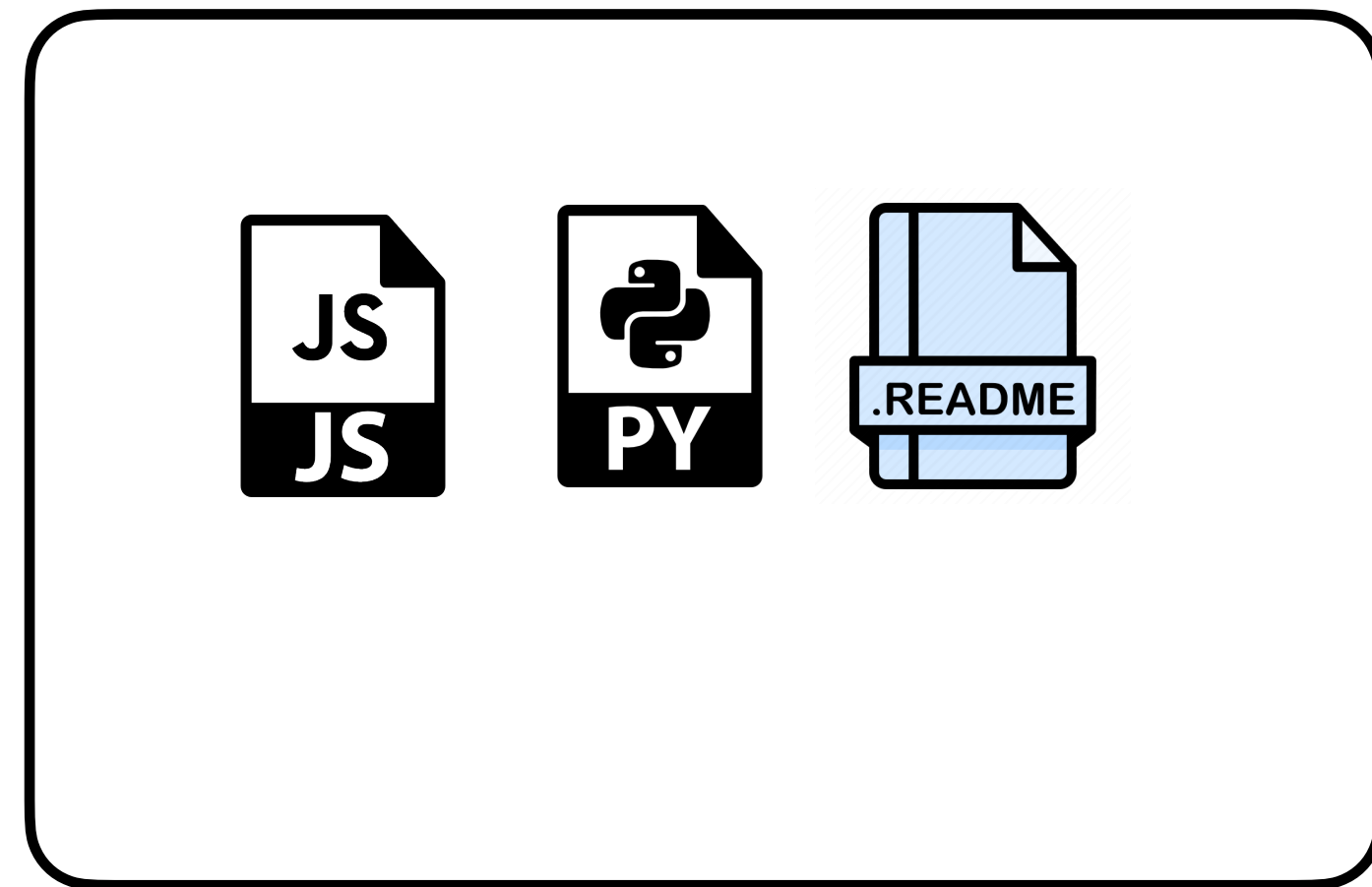
General git workflow

Working directory



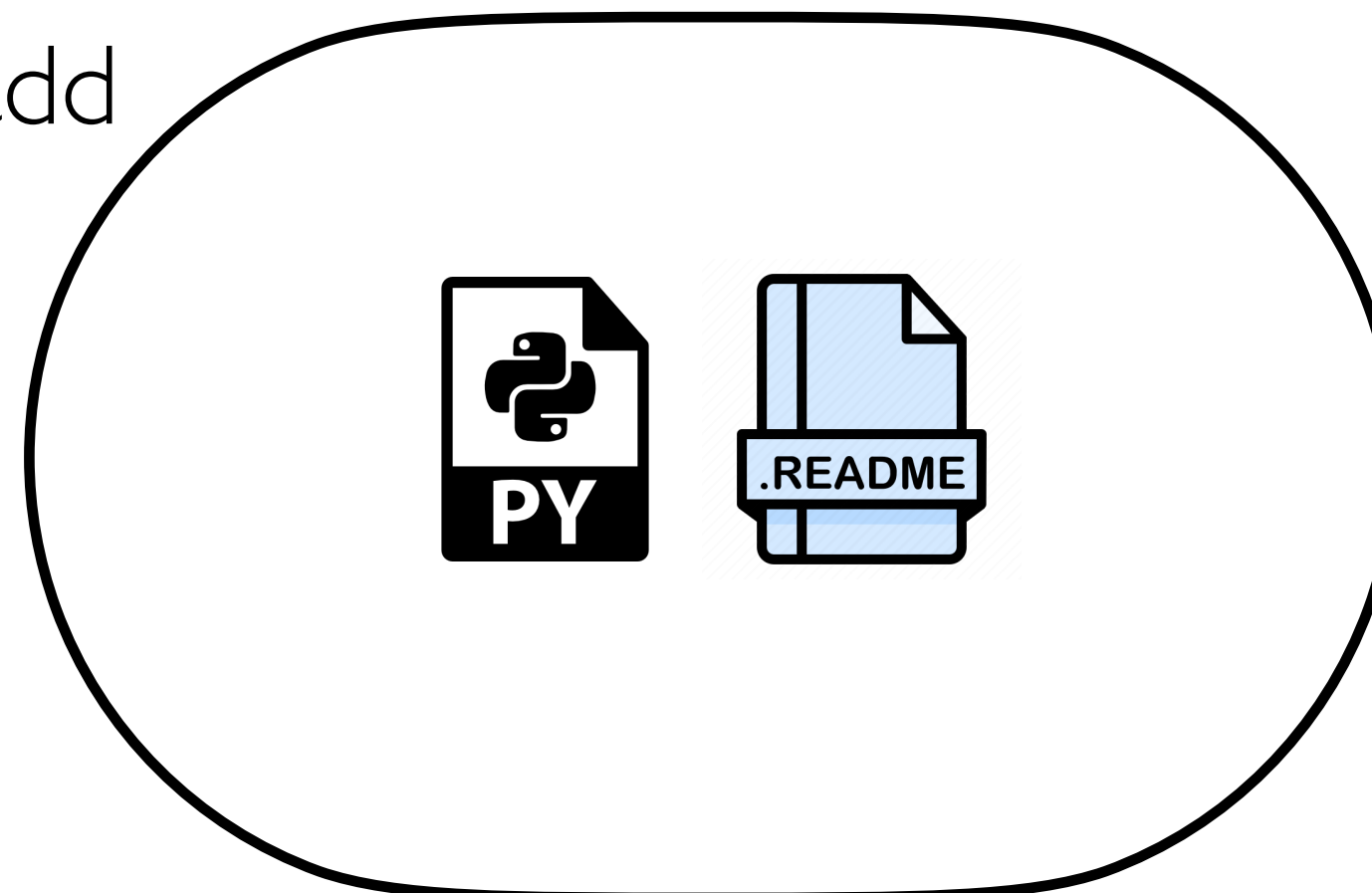
General git workflow

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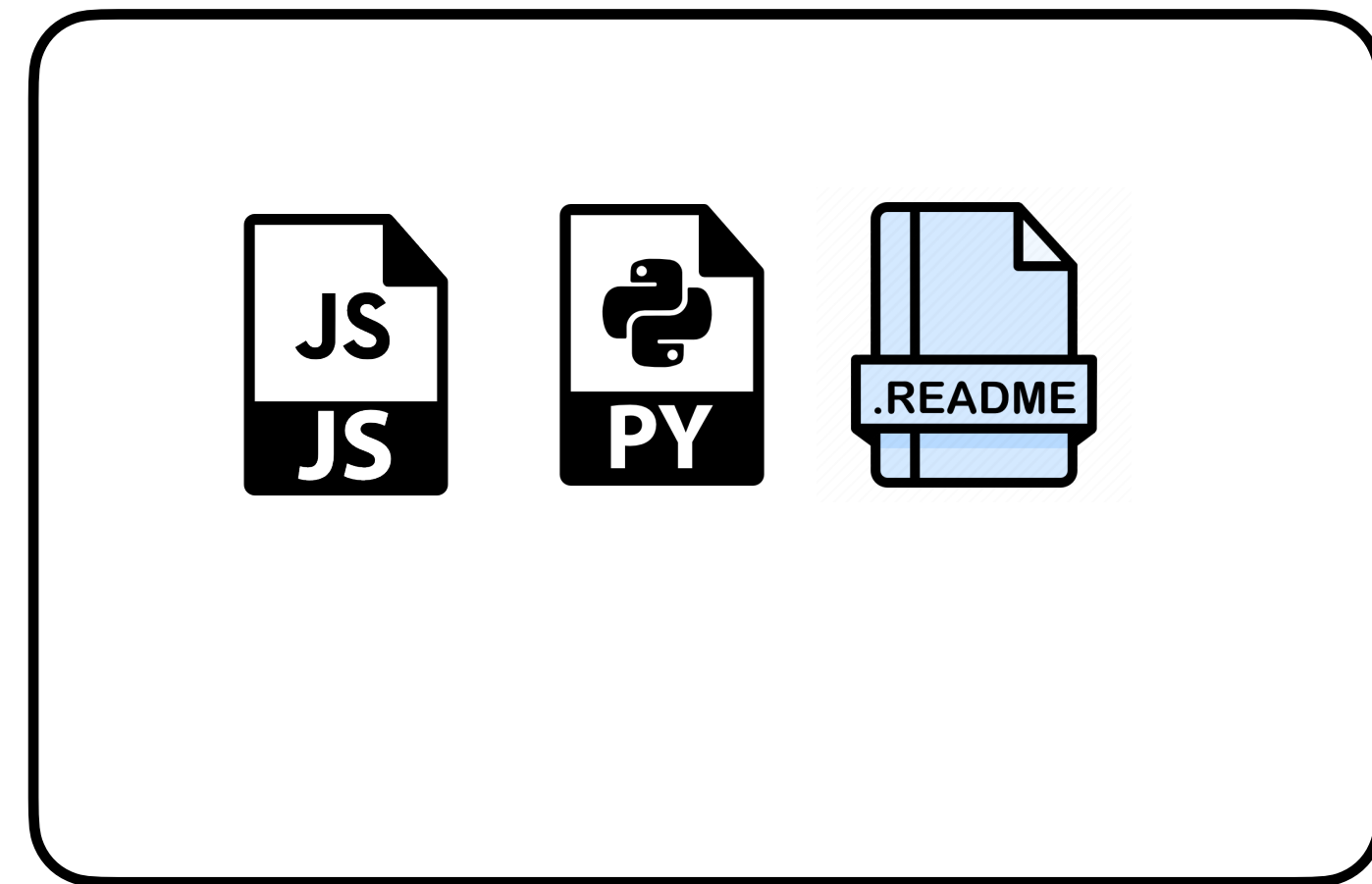
Git add

Staging or Index (old name)



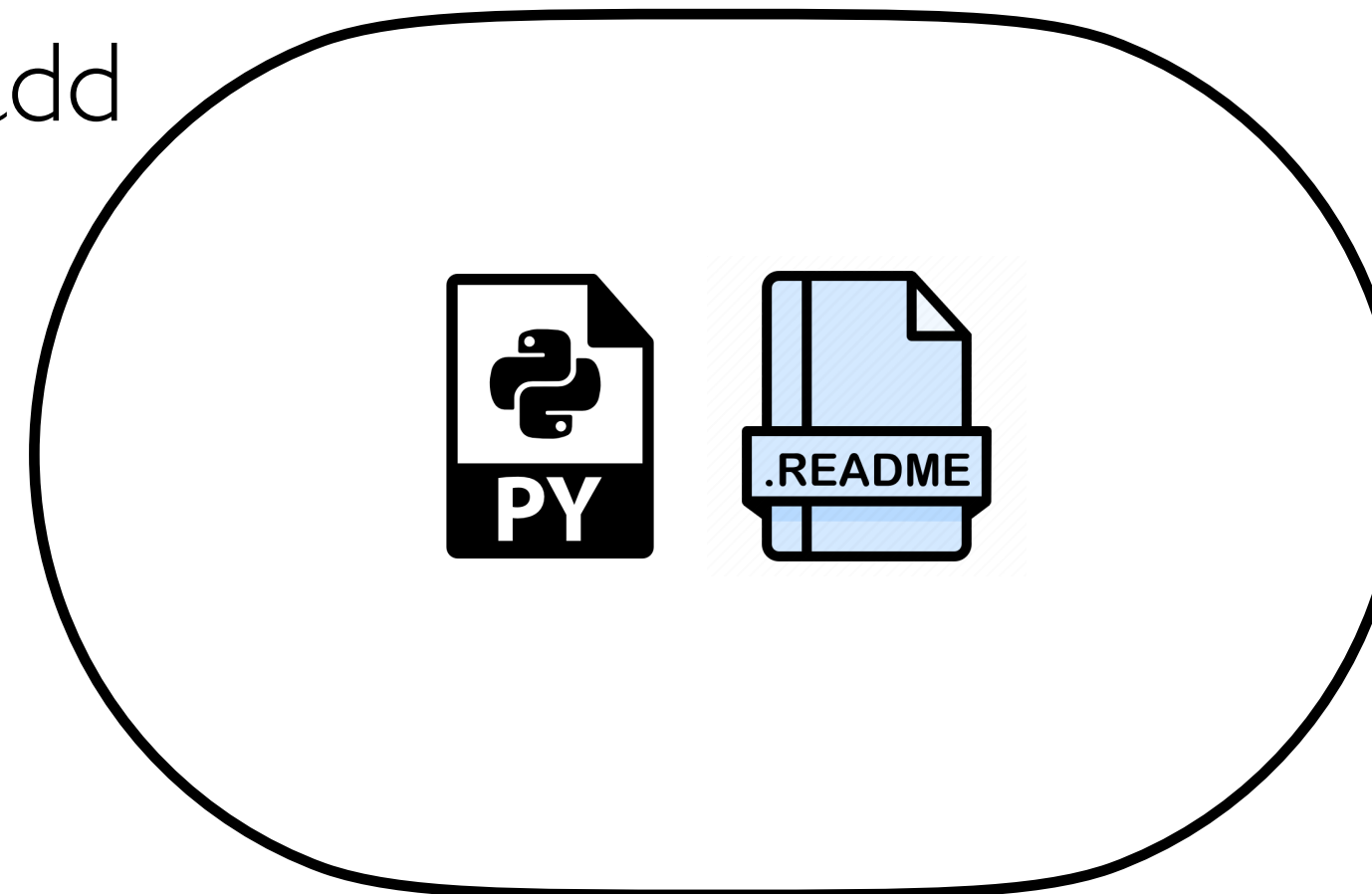
General git workflow

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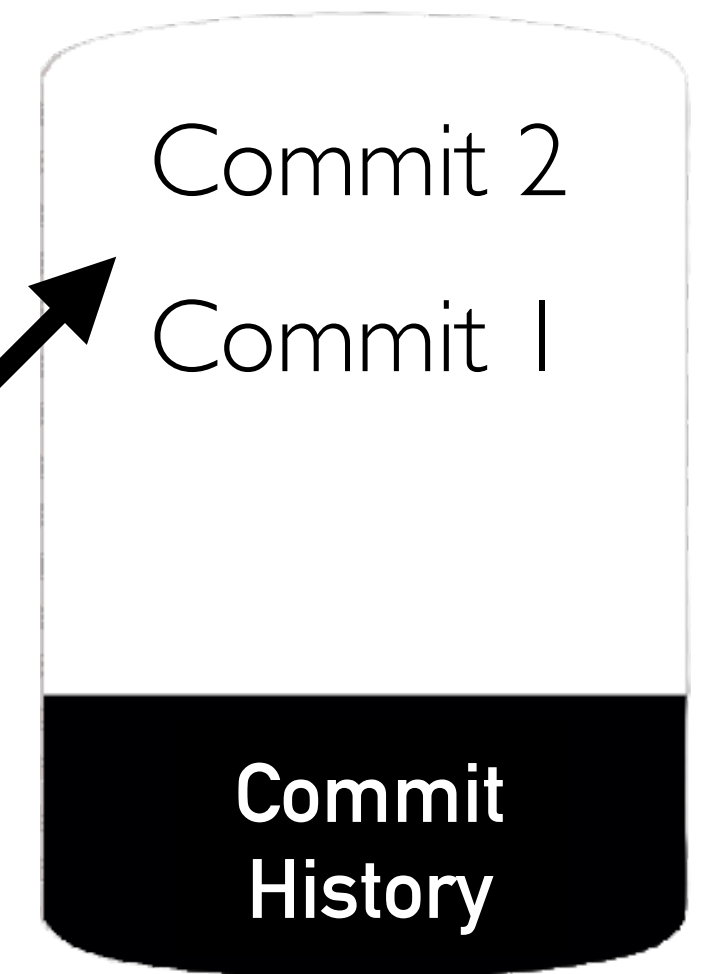


Git add

Staging or Index (old name)

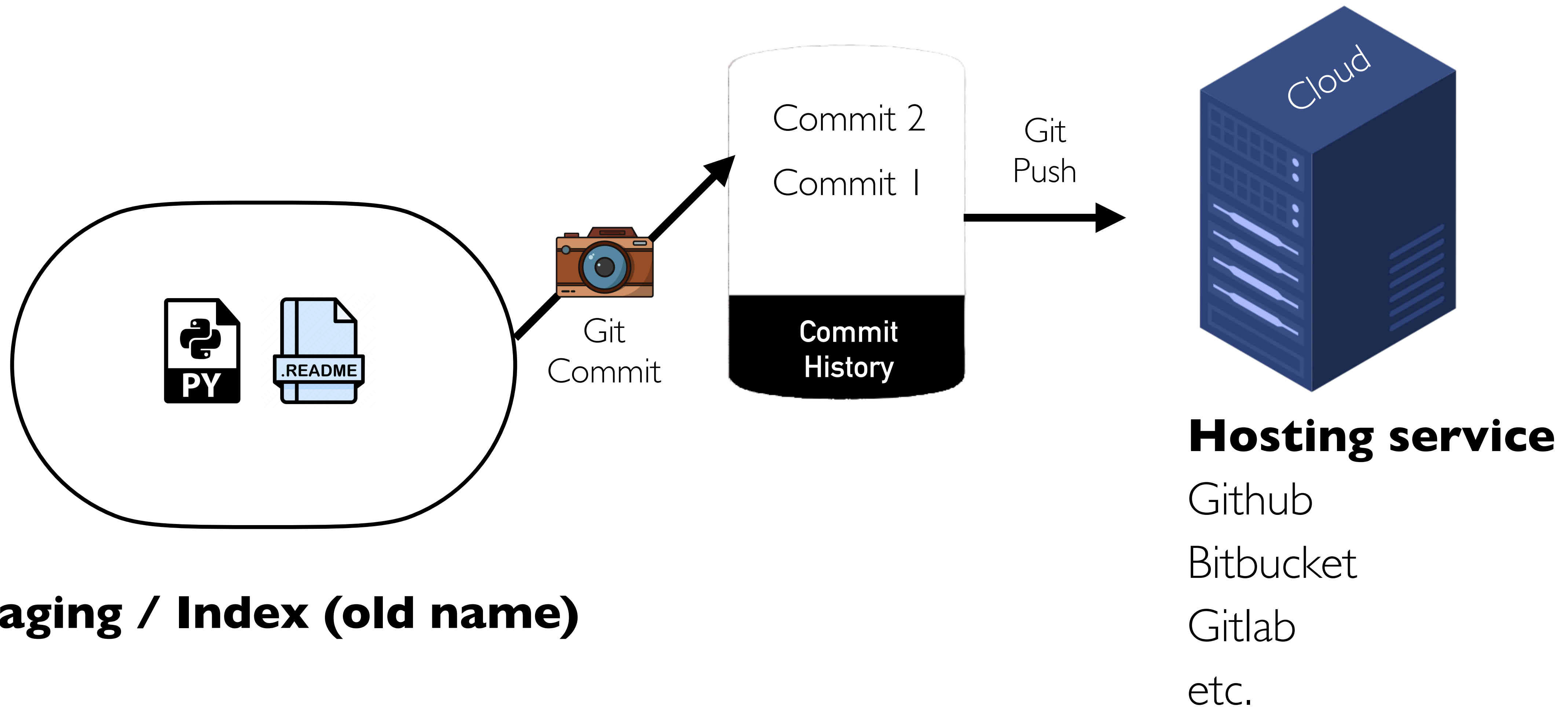


Git
Commit



Ps. Git doesn't save the same files for each commit. Files are compressed and saved uniquely

General git workflow



Git installation

- ◆ Installation instructions : <https://github.com/git-guides/install-git>
- ◆ For windows use Git BASH

```
$ git version  
git version ****
```

Git configuration

Cf. <https://git-scm.com/book/sv/v2/Customizing-Git-Git-Configuration>

```
$ git config [config level] [config name] [config value]
```

◆ [config levels] :

--local : (default) applies to the local repository, config is found in .git/config

--global : applies to the OS user repositories, config is found in ~/.gitconfig

--system : applies to the all OS users, config is found in /etc/gitconfig

◆ [config Name] :

user : email | name

core : editor

alias : git command

```
$ git config --global user.name 'amine'
```

```
$ git config --global user.email 'youremail@domain.com'
```

```
$ git config --local alias.st status -s
```

```
$ git config --global alias.lg 'log --oneline --graph --all'
```

```
$ git config --global -e #To see the current configuration
```

Bash commandes reminder

\$ cd

\$ ls -A

\$ pwd

\$ chmod

\$ grep

\$ mv

\$ cp

\$ rm -r

\$ mkdir

\$ touch

\$ cat

\$ diff

\$ echo

\$ man

\$ history

\$ clear

\$ locate

\$ open .

\$ exit

\$ kill

\$ head

\$ tail

\$ ps

Initialise git

```
$ git init .
```

```
$ mkdir "hello git" && cd "$_"
```

```
$ git init .
```

```
$ ls -a
```

```
.  ..  .git
```

Add file(s) to the staging area

```
$ git add <file>  
$ git ls-files #like ls on stage
```

- ◆ <file> can be :
 - ▶ String : File name
 - ▶ • : all file within directory (Not recommended)
 - ▶ * • [extension]

```
$ touch main.py  
$ echo "print('hello world')" > main.py  
$ cat main.py  
$ git add main.py
```

Make a commit (Take snapshot)

```
$ git commit -m "your commit message here"
```

```
$ git commit -m "Initial commit."
```

Change (last) commit message

```
$ git commit --amend -m "your new commit message"
```

You can commit and add on the same time (adds files already tracked to the staging)

```
$ git commit -am "your new commit message"
```

Git log

Log history of commits

```
$ git log  
$ git log --oneline  
$ git log --oneline --reverse #start from the first commit
```

Commit hash

Current branch

Commit message

```
[macbook-pro-de-amine:hello git amine$ git log  
commit 61ad5ba39f8dccdae64b5b1a1825bbb2138cc0b6 (HEAD -> main)  
Author: AmineFrj <ferdjaouiamine@gmail.com>  
Date: Mon Nov 28 14:12:48 2022 +0100  
  
    Rename main.py  
  
commit fb90b43d2d78c2c709da49afe8e65e7d40e6d69f  
Author: AmineFrj <ferdjaouiamine@gmail.com>  
Date: Mon Nov 28 14:06:22 2022 +0100  
  
    add readme
```

Git show & HEAD

View commit details

```
$ git show <hash>
$ git show HEAD~N #The N-th commit before head
$ git show HEAD~N:<filename>
$ git ls-tree HEAD~N #like ls -a on certain commit
```

Head is a pointer to the a branch (branche's latest commit)*

*The HEAD could also be pointing to another commit and in that case it is called *Detached HEAD*